Every exorcism has a price: ghost removal as survival conditioning, from Maldacena's boundary condition to the pole sheet

Jonathan Torkelson
Embeddetech, Inc., Tulsa, Oklahoma, USA
jtorkelson@embeddetech.com

July 19, 2026

Abstract

Draft — not for circulation.

A higher-derivative (ghost) theory can be made ghost-free by a choice: a boundary condition, a contour, a norm. The companion papers of this series established a currency in which such choices can be priced: a quantum evolution admits an ordinary ("beable") sub-stochastic process whenever its survival amplitude stays bounded—divisor negativity then marking indivisibility rather than forcing conditioning—and only when the amplitude grows without bound does the dynamics survive solely as a *survival-conditioned* process—which, for the coupled ghost realizations priced here, is the terminal-indivisibility class, with a strictly negative generator floor (unboundedness alone forces conditioning but not terminality: a decoupled growing mode stays divisible). This paper applies that currency to ghost *removal*. The thesis: every standard exorcism—boundary conditions that project the ghost out, and propagator prescriptions that redefine its pole—is survival conditioning in disguise, or buys its way out in another trilemma currency; for the boundary-condition removals and the Feynman route that price is computed here. Three results. (i) *Maldacena's future boundary condition* (conformal $\to$ Einstein gravity) is, mode by mode, a Gaussian effect $\mathcal{M}$ (regularized E_σ) on the Ostrogradski velocity variable of a detuned Pais–Uhlenbeck proxy whose detuning δ represents the de Sitter $\Delta = 3/\Delta = 2$ splitting. Its price sheet is closed form: survival $p = [1 + 1/(2\sigma^2\delta)]^{-1/2} \xrightarrow{\sigma\to 0} \sigma\sqrt{2\delta}$, and Einstein-vacuum fidelity of the delivered state $F = 2\sqrt{\alpha\omega_1}/(\alpha+\omega_1) \simeq 2\sqrt{\omega_2\delta}$ (with $\alpha = \omega_1\omega_2\delta$, in units $\omega_1 = 1$). Both collapse at the conformal (Minkowski, $\delta \to 0$) point—the known flat-space obstruction, now priced; the convention-free content is the survival collapse and the delivered state's absolute $\delta^{-1/2}$ width divergence, the fidelity reading the latter against a fixed Einstein-vacuum reference. (ii) The *Hell–Lüst–Zoupanos two-condition variant* $\mathcal{M}'$ fixes a canonically conjugate pair ($\dot{h}$ and the conjugate combination $\ddot{h} + \omega_1^2 h$, $[q_2, O_2] = \mathrm{i}$), so $\mathcal{M}'$ is Heisenberg-limited: it pays the same leading $\sigma\sqrt{2\delta}$ survival power plus an $O(1)$ ordering/phase-space-cell surcharge, and what the surcharge buys is not a cheaper exorcism but a more general one—the full two-parameter Einstein solution with no in-state restriction. The $\delta \to 0$ collapse is not evaded. (iii) *The pole sheet is readable in the divisor*: by Bochner's theorem a second-sheet resonance (positive spectral measure) has $|A(t)| \le 1$ and survives as an *unconditioned* sub-stochastic process (its divisor negativity transient when the amplitude is eventually monotone decreasing), while a first-sheet ghost pair (a discrete complex-conjugate pole pair off the real axis) has unbounded $|A(t)|$ in states overlapping the growing leg, so conditioning is *mandatory*; for the dimer realization it is terminal, with the generator floor of the first companion paper. Buoninfante's

multi-particle masking law $\sinh^2(\Gamma t/2)$ is exactly the time dependence of the cross-population of the terminal-class (conditioning-mandatory) dimer (with $\widetilde{k} = \Gamma/2$), up to the normalization $(\kappa/\widetilde{k})^2$, and his ghost-prescription taxonomy (Feynman ghost / anti-Feynman / fakeon) maps onto the trilemma currencies of the series: sample space, causal order, silence. Unstable particles survive unconditioned; ghosts survive only conditioned—and every way of pretending otherwise pays, measurably, in one currency or another.

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *"Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?"* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every result is stated with the proof it actually has—closed-form derivations where they exist, and numerically established claims labeled as such—and every numerical claim is reproducible from the public scripts (see Code availability); the originating conversation, project briefing, and full research notebook are preserved in the program's hub repository, https://github.com/embeddetech/ghosts-program.

Contents

1 Introduction **4**

2 The currency: divisors, conditioning, and the detuned Pais–Uhlenbeck mode **7**
2.1 Divisors and the terminal class 7
2.2 The detuned Pais–Uhlenbeck proxy 7

3 Maldacena's exorcism, priced **9**
3.1 What is the effect and what is the state preparation 9
3.2 The survival price 9
3.3 The delivered state 10
3.4 The price sheet and the conformal collapse 10

4 The Hell–Lüst–Zoupanos variant: a Heisenberg-limited exorcism **11**
4.1 Two future conditions, one conjugate pair 11
4.2 The price of $\mathcal{M}'$ 13
4.3 What the surcharge buys 14

5 The pole sheet is readable in the divisor **15**
5.1 The Bochner discriminant 15
5.2 The masking law is the terminal-class dimer's cross-population 17

6 A taxonomy of exorcisms and their currencies **17**

7 Relation to final-state post-selection **18**

8 Limitations **19**

9 Conclusion **20**

Author contributions and use of AI **20**

Code availability **21**

1 Introduction

Ghosts are removed, in the literature, in two broad ways. The first is by *boundary condition*: one imposes conditions at a future (or asymptotic) boundary that the ghost modes fail, so that only the healthy sector survives. The paradigm is Maldacena's derivation of Einstein gravity from conformal gravity [4]: the fourth-order graviton equation has four solutions per mode, a positive-frequency condition in the far past halves them, and a Neumann condition $\dot{g} = 0$ at the future boundary halves them again, leaving exactly the Einstein graviton; at tree level the conformal-gravity wavefunctional evaluated on Neumann data reproduces the (renormalized) Bunch–Davies wavefunctional of Einstein gravity. A recent variant by Hell, Lüst, and Zoupanos [5] keeps the Neumann condition but adds a second future condition on $\ddot{h}$ and drops the in-state restriction, aiming at the *general* Einstein solution. The second removal is by *prescription*: one keeps the ghost in the propagator but redefines how its pole is handled—the Feynman ghost,

the anti-Feynman (advanced-arrow) ghost, and Anselmi's fakeon are the three canonical choices, recently organized into a sharp taxonomy by Buoninfante [6, 7, 8] and disputed by Anselmi, Briscese, Calcagni, and Modesto [9], with the Donoghue–Menezes "Merlin mode" treatment [10] occupying a related corner.

The companion papers of this series supply a currency in which such removals can be priced rather than merely asserted. The foundational paper [1] shows that a non-unitary or higher-derivative quantum evolution admits an ordinary ("beable") sub-stochastic process—definite configurations, nonnegative transition probabilities—whenever its survival amplitude stays bounded; the two-time *divisor* $M(t_2, t_1) = \Gamma(t_2)\Gamma(t_1)^{-1}$ of its transition matrices staying entrywise nonnegative on $[t_1, t_2]$ marks the process *divisible* there, and divisor negativity marks indivisibility, not non-existence. When the negativity is only transient, the process is indivisible but ordinary. Only when the amplitude grows without bound—so that no unconditioned sub-stochastic process exists—does the dynamics survive solely as a *survival-conditioned* process, which for the coupled ghost realizations treated here is the terminal-indivisibility class (permanent divisor negativity past a critical time, the conditioned generator pinned at a strictly negative floor), in which every probability is conditioned on an ensemble that is itself being consumed. (Unboundedness by itself forces the conditioning but not the terminality—a decoupled mode that merely grows stays divisible; the terminal floor is a feature of the coupling.) The trilemma companion [3] then prices the field-theory options: a coupled-ghost theory keeps at most two of {coupled ghosts, empirical Lorentz invariance, definite configurations} at tree level, and [2] makes the Lorentz cost concrete through the boost-orbit measure of ghost vacuum decay.

The Ostrogradski ghost of a higher-derivative theory is, on this view, not a calculational error to be deleted but a structural feature that must be paid for. This paper's thesis is that ghost removal itself belongs on that price sheet:

> *Every standard exorcism is survival conditioning in disguise, or buys its way out in another trilemma currency; for the boundary-condition removals and the Feynman route that price is computed here.*

For boundary-condition removals the statement is nearly literal: a future boundary condition that the prepared state does not already satisfy is a quantum *effect*—a post-selection—and the process that results is survival-conditioned by construction. What has been missing is the price. For prescription removals the statement is structural: the divisor reads the pole sheet, and the first-sheet ghost pole forces conditioning while the second-sheet resonance does not; the prescriptions that avoid conditioning pay instead in the other trilemma currencies.

Concretely, we establish three groups of results.

(i) Maldacena's boundary condition, priced (Section 3). We translate the per-mode fourth-order graviton system into a detuned Pais–Uhlenbeck (PU) oscillator with frequencies ω_1 and $\omega_2 = \omega_1 - \delta$, the detuning δ standing for the de Sitter splitting of the $\Delta = 3$ graviton from its $\Delta = 2$ negative-norm partner; Minkowski is the conformal point $\delta \to 0$, where the pair degenerates into a Jordan block. In Ostrogradski variables $q_1 = x$, $q_2 = \dot{x}$ the Neumann condition is a condition on a *configuration* variable, so its regularized form is a bona fide multiplication effect $E_\sigma = \exp(-q_2^2/2\sigma^2)$, $0 \le E_\sigma \le 1$; Maldacena's positive-frequency condition is the state preparation and is deliberately *not* counted in the price. On the Dirac-normalizable vacuum of the declared quantization the price sheet is closed form:

$$p(\sigma,\delta) = \left[1 + \tfrac{1}{2\sigma^2\delta}\right]^{-1/2} \xrightarrow{\sigma\to 0} \sigma\sqrt{2\delta}, \qquad F = \frac{2\sqrt{\alpha\,\omega_1}}{\alpha+\omega_1} \simeq 2\sqrt{\omega_2\delta}, \quad \alpha = \omega_1\omega_2\delta \quad (\omega_1 = 1),$$

where p is the survival probability of the effect and F the fidelity of the delivered (conditioned) graviton state to the Einstein vacuum (against a fixed unit-mass reference; the convention-free content is the survival collapse and the delivered state's absolute width divergence). Both vanish as $\delta \to 0$: the exorcism works in de Sitter, where $\delta = O(1)$, and its price diverges exactly at the Minkowski point where Hell–Lüst–Zoupanos found the flat-space version fails. The known qualitative obstruction becomes a quantitative collapse of the price sheet.

(ii) The Hell–Lüst–Zoupanos variant is Heisenberg-limited (Section 4). HLZ impose two future conditions—Neumann $\dot{h}|_0 = 0$ (their Eq. (30)) and a curvature-matching condition on $\ddot{h}$ (their Eq. (59) in covariant form; per Fourier mode their Eq. (64), $\ddot{h}_k|_0 = k^2 h_k|_0$)—and no in-state restriction, delivering the general two-parameter Einstein-Λ tensor mode (their Eq. (67)). In the Ostrogradski frame the second condition constrains $O_2 = p_2 + \omega_1^2 q_1$, and p_2 is the momentum *conjugate* to the q_2 that Neumann fixes: $[q_2, O_2] = \mathrm{i}$. The two-condition effect $\mathcal{M}'$ is therefore a joint position–momentum (Heisenberg-limited) effect. Its *Weyl-ordered* survival factorizes at the Gaussian level ($\mathrm{cov}(q_2, O_2) = 0$ exactly), the second condition is asymptotically free ($\mathrm{Var}(O_2) = \frac{\delta}{2}\frac{\omega_1+\omega_2}{\omega_2} \to 0$; the ghost-killing sign is also the finite-variance one, the opposite sign giving $\mathrm{Var}(O_2) \to \infty$, Prop. 4.1), and both natural operator orderings then give the same leading power $p_{\mathcal{M}'} \sim \sigma\sqrt{2\delta}$, differing only by an $O(1)$ phase-space-cell factor. The surcharge buys generality, not cheapness; the conformal collapse is not evaded.

(iii) The pole sheet is readable in the divisor (Sections 5–6). Buoninfante's trilogy [6, 7, 8] isolates the sharp spectral difference between an ordinary unstable particle and a ghost with the same complex-conjugate pole pair $m^2 \pm \mathrm{i}m\Gamma$: for the unstable particle the pair sits on the *second* sheet and the Källén–Lehmann measure is a positive density with sum rule $1 = C$ (all norm transfers to the continuum); for the ghost the pair sits on the *first* sheet and the spectral decomposition acquires discrete *negative-residue* complex-pole terms with sum rule $Z + Z^* = 1 + C$ (the one-particle component persists). We show this dichotomy is exactly the divisor dichotomy of [1]: a positive spectral measure makes the survival amplitude positive-definite in Bochner's sense, so $|A(t)| \le 1$ and an *unconditioned* sub-stochastic survival process exists, its divisor negativity transient when the amplitude is eventually monotone decreasing; the first-sheet pair makes $|A(t)|$ grow without bound, so no unconditioned process exists and the survival-conditioned dilation is mandatory—and, for the dimer realization, it lands in the terminal class, with the critical time and generator floor of [1] in closed form. Buoninfante's "anti-instability" masking law, the multi-particle overlap growing as $\sinh^2(\Gamma t/2)$, is shown to be *exactly* the time dependence of the first companion paper's terminal-class (conditioning-mandatory) dimer cross-population—the pre-conditioning growth that conditioning divides out—$|U_{12}(t)|^2 = (\kappa/\widetilde{k})^2 \sinh^2(\widetilde{k}t)$ with $\widetilde{k} = \Gamma/2$, up to the normalization $(\kappa/\widetilde{k})^2$: two bookkeepings of the same pole pair, one in the closed indefinite-metric currency, one in the beable currency. Finally the prescription taxonomy maps onto the trilemma: the Feynman ghost pays sample space (conditioning), the anti-Feynman ghost pays causal order (the Donoghue–Menezes corner—a connection this paper draws), and the fakeon pays silence (the state is excised). The live Anselmi–Buoninfante dispute over which prescription is "viable" is, in this reading, a currency dispute, and the trilemma prices both sides.

Section 7 scopes the claims against the final-state post-selection literature [11, 12]; Section 8 lists the limitations, all of which are substantive; Section 9 concludes. Throughout, statements proved in closed form are labeled propositions with proofs; statements established numerically with analytic structure are labeled as such in the proof text. All numbers are reproducible from

three short scripts (Code availability).

2 The currency: divisors, conditioning, and the detuned Pais–Uhlenbeck mode

2.1 Divisors and the terminal class

We recall the minimal apparatus of [1]. Let a finite-configuration dynamics be presented by transition matrices $\Gamma(t)$ from time 0 to time t (columns summing to at most one: a *sub-stochastic* process, the deficit being read as decay or, in the conditioned case, as failure to survive post-selection). The process is an ordinary (beable) process on the interval $[t_1, t_2]$ iff the *divisor*

$$M(t_2, t_1) \;=\; \Gamma(t_2)\,\Gamma(t_1)^{-1} \tag{1}$$

is entrywise nonnegative: M is then itself a transition matrix and the process can be continued through t_1 without memory of how it got there. (For a strictly sub-stochastic Γ one additionally needs M column-sub-stochastic; entrywise nonnegativity is the operative criterion, the normalization tracked separately, and the full statement is that of [1].) Transient divisor negativity (indivisibility) is compatible with an ordinary process on coarser time grids; the pathological class identified in [1] is *terminal* indivisibility: negativity that sets in at a finite t_c and never leaves, with the late-time conditioned generator pinned at a strictly negative off-diagonal floor. A unitary (or non-unitary) mode dynamics $U(t)$ that admits no sub-stochastic Γ at all—because $\|U(t)\| > 1$—survives in the beable currency only through the survival-conditioned dilation: normalize by the largest singular value,

$$\Gamma_{\mathrm{c}}(t) \;=\; \Big|\,U(t)/\sigma_{\max}\big(U(t)\big)\Big|^{2} \quad \text{(entrywise modulus squared)}, \tag{2}$$

which is the transition matrix of the process conditioned on surviving the growth. For the broken-$\mathcal{PT}$ dimer

$$H = \begin{pmatrix} \mathrm{i}a & \kappa \\ \kappa & -\mathrm{i}a \end{pmatrix}, \qquad a > \kappa, \qquad \widetilde{k} = \sqrt{a^2 - \kappa^2}, \tag{3}$$

the first companion paper derives the closed forms

$$t_c = \frac{1}{\widetilde{k}}\,\operatorname{arcsinh}\!\Big(\frac{\widetilde{k}}{\kappa\sqrt{2}}\Big), \qquad \text{generator floor} \;=\; -(a + \widetilde{k}), \tag{4}$$

both of which are re-verified numerically in Section 5. Because $H^2 = -\widetilde{k}^2\mathbf{1}$ (the σ_z and σ_x parts anticommute),

$$U(t) = \mathrm{e}^{-\mathrm{i}Ht} = \cosh(\widetilde{k}t)\,\mathbf{1} - \mathrm{i}\,\frac{\sinh(\widetilde{k}t)}{\widetilde{k}}\,H, \qquad |U_{12}(t)|^2 = \frac{\kappa^2}{\widetilde{k}^2}\,\sinh^2(\widetilde{k}t), \tag{5}$$

an exact identity we will need in Section 5.2.

2.2 The detuned Pais–Uhlenbeck proxy

Per mode, the linearized fourth-order graviton dynamics of conformal gravity is a Pais–Uhlenbeck oscillator [13]. We work with the detuned PU Lagrangian

$$L = \tfrac{1}{2}\big[\ddot{x}^2 - (\omega_1^2 + \omega_2^2)\,\dot{x}^2 + \omega_1^2\omega_2^2\,x^2\big], \qquad \omega_2 = \omega_1 - \delta, \qquad 0 < \delta < \omega_1, \tag{6}$$

with solutions $e^{\pm i\omega_1 t}, e^{\pm i\omega_2 t}$. We set $\omega_1 = 1$ throughout the numerics. The detuning δ is our proxy dial for the de Sitter splitting: in Maldacena's analysis [4] the surviving graviton is the $\Delta = 3$ mode and its removed partner is the negative-norm $\Delta = 2$ mode, split by the Hubble scale; the Minkowski limit $H_\Lambda \to 0$ merges the pair into the degenerate (t-growing, Jordan-block) conformal mode, i.e. $\delta \to 0$. The proxy is a declared, mode-level translation—we do not derive it from the conformal-gravity path integral, and Section 8 returns to this point.

Ostrogradski variables: $q_1 = x$, $q_2 = \dot{x}$, with conjugate momenta p_1, p_2 ($p_2 = \ddot{x}$ on shell), and

$$H = p_1 q_2 + \tfrac{1}{2} p_2^2 + \tfrac{1}{2}(\omega_1^2 + \omega_2^2)\, q_2^2 - \tfrac{1}{2}\, \omega_1^2 \omega_2^2\, q_1^2. \tag{7}$$

Both q_1 and q_2 are *configuration* variables: the beable sample space of this mode is the two-slice memory $(x, \dot{x})$, exactly as in the companion papers. H is Hermitian on $L^2(\mathbb{R}^2)$ and unbounded below (Ostrogradski); its normalizable spectrum in this quantization is $E_{n_1 n_2} = \omega_1(n_1 + \frac{1}{2}) - \omega_2(n_2 + \frac{1}{2})$.

Proposition 2.1 (The Dirac-normalizable vacuum and its exceptional-point degeneration)**.** *The Gaussian ansatz* $\Psi_0(q_1, q_2) = \exp\!\big[-\tfrac{1}{2}(\alpha q_1^2 + b q_2^2) - \mathrm{i}c\, q_1 q_2\big]$ *solves* $H\Psi_0 = E_0 \Psi_0$ *exactly, with*

$$c = \omega_1 \omega_2, \qquad b = \delta, \qquad \alpha = \omega_1 \omega_2\, \delta, \qquad E_0 = \tfrac{\delta}{2}. \tag{8}$$

Hence $|\Psi_0|^2 = \exp(-\omega_1 \omega_2 \delta\, q_1^2 - \delta\, q_2^2)$*: the state is normalizable for every* $\delta > 0$*, and its widths in* both *configuration variables diverge as* $\delta^{-1/2}$ *at the conformal point. The root* $c = -\omega_1\omega_2$*,* $b = \omega_1 + \omega_2$ *has* $\alpha = -\omega_1\omega_2(\omega_1 + \omega_2) < 0$ *and is normalizable only under contour rotation (the Bender–Mannheim quantization [14]); it is not used here.*

Proof. Substituting the ansatz into (7) with $p_1 = -\mathrm{i}\partial_{q_1}$, $p_2 = -\mathrm{i}\partial_{q_2}$ and collecting monomials gives three coefficient equations and the eigenvalue:

$$q_1^2: \quad \tfrac{c^2}{2} = \tfrac{\omega_1^2\omega_2^2}{2}; \qquad q_2^2: \quad \tfrac{b^2}{2} + c = \tfrac{\omega_1^2+\omega_2^2}{2}; \qquad q_1 q_2: \quad \alpha = bc; \qquad E_0 = \tfrac{b}{2}.$$

The first fixes $c = \pm\omega_1\omega_2$; with $c = +\omega_1\omega_2$ the second gives $b^2 = (\omega_1 - \omega_2)^2$, and the normalizable root is $b = \delta$, whence $\alpha = \omega_1\omega_2\delta$ and $E_0 = \delta/2$. With $c = -\omega_1\omega_2$ one finds $b = \omega_1 + \omega_2$, $\alpha < 0$. The three independent coefficient equations were also confirmed to machine precision, and the eigenvalue on a 1201^2 grid at $\delta = 0.5$ to $E_0 = 0.249998$ vs. 0.25 (the script noted in Code availability). □

The $\delta^{-1/2}$ width divergence is the exceptional-point degeneration of the equal-frequency limit made quantitative at the level of the state: as the two frequencies merge, the normalizable vacuum flattens in both beables at the same rate. Everything priced below inherits this degeneration.

Declared quantization (and why it must be declared). The PU mode at unequal frequencies admits inequivalent quantizations: the Dirac-normalizable realization above (positive norm, energy unbounded below—the "Hermitian ghost" realization, which is the one carrying a beable sample space, cf. [3]), and the contour-rotated Bender–Mannheim realization (bounded energy, indefinite or redefined norm). The price computed in this paper is a statement about the first. This is the honest order of operations: one must choose the quantization *before* pricing the effect, and the answer may depend on the choice. We state it once and do not hide it.

3 Maldacena's exorcism, priced

3.1 What is the effect and what is the state preparation

Maldacena's argument [4] has two moving parts per graviton mode. First, positive frequency in the far past: this selects the Bunch–Davies-type in-state. Second, the Neumann condition $\dot{g} = 0$ at the future boundary: this kills the $\Delta = 2$ partner and leaves the Einstein graviton, with $\Psi_{\text{Conformal}}[g, \dot{g} = 0] \propto \Psi_{\text{BD-ren}}[g]$ at tree level. In post-selection language the two parts are *not* on the same footing, and pricing them as if they were would double-count:

- the positive-frequency condition is *state preparation* (it fixes ρ_{in}, here proxied by Ψ_0 of Proposition 2.1);
- the Neumann condition is the *effect*: a condition imposed at a later boundary that the prepared state does not already satisfy.

Only the second carries a survival price, and only the second is what we price.

The sharp Neumann effect "$q_2 = 0$" is improper (a δ-normalized projection, with survival probability strictly zero against any absolutely continuous distribution). We therefore price the regularized Gaussian family

$$E_\sigma = \exp\Big(-\frac{q_2^2}{2\sigma^2}\Big), \qquad 0 \le E_\sigma \le 1, \qquad K_\sigma = E_\sigma^{1/2}, \tag{9}$$

a bona fide multiplication effect on the configuration variable $q_2 = \dot{x}$ (this is the point of the Ostrogradski frame: the velocity is a beable, so the Neumann condition is a position-type measurement, not a momentum smearing), and control the $\sigma \to 0$ limit explicitly. The determinant of E_σ on the relevant Gaussian sector and the survival probability vanish together, so the regularization introduces no spurious hierarchy; all statements below are made at fixed σ and then in the stated $\sigma \to 0$ asymptotics.

3.2 The survival price

Proposition 3.1 (Survival)**.** *On the vacuum of Proposition 2.1,*

$$p(\sigma, \delta) = \frac{\langle \Psi_0 | E_\sigma | \Psi_0 \rangle}{\langle \Psi_0 | \Psi_0 \rangle} = \Big[1 + \frac{1}{2\sigma^2 \delta}\Big]^{-1/2} \xrightarrow{\sigma \to 0} \sigma\sqrt{2\delta}\,. \tag{10}$$

Proof. The q_2-marginal of $|\Psi_0|^2$ is $\propto \mathrm{e}^{-\delta q_2^2}$ (the q_1 integral factors), so

$$p = \frac{\int \mathrm{e}^{-\delta q_2^2} \mathrm{e}^{-q_2^2/2\sigma^2}\, \mathrm{d}q_2}{\int \mathrm{e}^{-\delta q_2^2}\, \mathrm{d}q_2} = \sqrt{\frac{\delta}{\delta + 1/2\sigma^2}} = \Big[1 + \tfrac{1}{2\sigma^2\delta}\Big]^{-1/2}.$$

The grid evaluation agrees with this closed form to six digits at all tested (σ, δ). □

The physics is in the δ-dependence, and it is transparent: the vacuum's spread in the very variable the exorcism constrains grows as $\delta^{-1/2}$ (Proposition 2.1), so the fraction of the ensemble that survives the constraint shrinks as $\sqrt{\delta}$. In de Sitter, where the $\Delta = 3/\Delta = 2$ splitting is order one, the price is order σ—the ordinary cost of any finite-resolution post-selection. As the cosmological constant is switched off the exorcism becomes arbitrarily expensive.

3.3 The delivered state

Removing the ghost is only half of the advertisement; the other half is that what survives is *Einstein* gravity. The proxy for that claim is the fidelity of the conditioned state to the Einstein-mode vacuum $\phi(q_1) = (\omega_1/\pi)^{1/4} \mathrm{e}^{-\omega_1 q_1^2/2}$.

Proposition 3.2 (Delivery)**.** *As $\sigma \to 0$ the conditioned reduced state on q_1 is the pure Gaussian $\psi_c(q_1) \propto \Psi_0(q_1, 0) = \mathrm{e}^{-\alpha q_1^2/2}$ with $\alpha = \omega_1\omega_2\delta$, and its fidelity to the Einstein vacuum is*

$$F = \left|\langle\phi|\psi_c\rangle\right|^2 = \frac{2\sqrt{\alpha\,\omega_1}}{\alpha + \omega_1} \;=\; \frac{2\sqrt{\omega_2\delta}}{1+\omega_2\delta}\Big|_{\omega_1=1} \;\simeq\; 2\sqrt{\omega_2\delta} \quad (\delta \to 0). \tag{11}$$

Proof. As $\sigma \to 0$ the Kraus operator K_σ concentrates the q_2 dependence on $q_2 = 0$; since $\Psi_0(q_1, q_2)$ is Gaussian, the conditioned q_1 slice is $\mathrm{e}^{-\alpha q_1^2/2}$ (the phase $\mathrm{e}^{-\mathrm{i}cq_1q_2}$ drops on the slice). The overlap of two centered Gaussians of widths $\alpha^{-1/2}, \omega_1^{-1/2}$ gives (11) by a one-line integral. Numerically, the log–log slope of $F(\delta)$ over $\delta \in [8\times10^{-4}, 2\times10^{-2}]$ is 0.491 (predicted 1/2), and the finite-σ *mixed* conditioned state at $(\delta, \sigma) = (0.1, 0.05)$ gives $F = 0.5500$ against the $\sigma \to 0$ closed form 0.5505: the pure-slice limit is reached smoothly. □

The delivered graviton is a Gaussian of width $\alpha^{-1/2} \sim (\omega_2\delta)^{-1/2}$, not the Einstein-vacuum width $\omega_1^{-1/2}$: the exorcised state remembers the exceptional-point flattening of the parent vacuum. At $\delta = O(1)$ (de Sitter) the fidelity is order one ($F = 0.80$ at $\delta = 0.5$); at the conformal point it collapses like $\sqrt{\delta}$. This fidelity is taken against a *fixed* external Einstein vacuum ϕ of δ-independent width $\omega_1^{-1/2}$ (the healthy graviton at unit mass)—a declared convention. The proxy's *own* healthy normal mode co-degenerates with the delivered state ($\mathrm{Var}(y_1) = 1/(2\omega_1\delta(\omega_1+\omega_2)) \to \infty$ as $\delta \to 0$), so measured against that co-scaling reference the fidelity would stay $O(1)$ (tending to $2\sqrt{2}/3$). The $\sqrt{\delta}$ collapse is therefore the delivered state's *absolute* width divergence $\alpha^{-1/2} \sim \delta^{-1/2}$ read against a fixed ruler, not a reference-independent fact; the convention-free content is (a) the survival collapse $p \sim \sigma\sqrt{2\delta}$ and (b) that absolute width divergence. (At $\delta = 0$ neither state is normalizable, so F is a ratio of two diverging widths there.)

3.4 The price sheet and the conformal collapse

δ	p (survival, $\sigma = 0.05$)	F (Einstein fidelity)
0.500	0.04994	0.8000
0.100	0.02236	0.5505
0.020	0.01000	0.2746
0.004	0.00447	0.1257

Table 1: The price sheet of Maldacena's exorcism on the detuned-PU proxy: survival probability of the regularized Neumann effect (Proposition 3.1) and Einstein-vacuum fidelity of the delivered state (Proposition 3.2), both closed form, both grid-verified. Both columns collapse at the conformal (Minkowski) point $\delta \to 0$; the survival collapse and the delivered state's *absolute* width divergence are convention-free, while the fidelity column reads that divergence against a *fixed* Einstein-vacuum width (text after Proposition 3.2).

Corollary 3.3 (Conformal collapse)**.** *$p\,F \simeq 2\sqrt{2}\,\sigma\,\omega_2^{1/2}\,\delta$ as $\delta \to 0$ at fixed σ: the joint figure of merit of the exorcism—how much ensemble survives times how Einstein-like the survivor is—vanishes* linearly *in the de Sitter splitting. (The F factor inherits the fixed-reference convention of Proposition 3.2; the survival factor p is convention-free.)*

This is the quantitative form of a failure that Hell, Lüst, and Zoupanos located qualitatively [5]. Their flat-space analysis finds that the Neumann-selected survivor in Minkowski is *not* an Einstein solution—a $t\,\mathrm{e}^{-\mathrm{i}kt}$ growing piece survives inside the selected state—while in de Sitter the survivor is Einstein but carries only half the Einstein data. Our diagnosis of the root cause: flat space is the degenerate PU point. At $\delta = 0$ the four-dimensional solution space does not split invariantly into graviton $\otimes$ partner (the mode matrix is a Jordan block), so writing the removal as $\mathcal{M} = \mathbf{1} \otimes E$ is singular exactly at $H_\Lambda = 0$; de Sitter works because the $\Delta = 3/\Delta = 2$ splitting resolves the degeneracy. The prediction that follows—the price should degenerate as $H_\Lambda \to 0$—is what Table 1 confirms: the divisor currency *detects* the flat-space obstruction as a price collapse, without being told about Jordan blocks. For the de Sitter over-selection (half the Einstein data), the post-selection reading is actually charitable: an effect is supposed to pin a state; the "missing" half of the classical data is not a bug of the exorcism-as-conditioning but of reading it as a classical solution map. What one *does* pay to recover the general Einstein solution is the subject of the next section.

4 The Hell–Lüst–Zoupanos variant: a Heisenberg-limited exorcism

4.1 Two future conditions, one conjugate pair

HLZ [5] replace Maldacena's (state-prep + one condition) scheme by two future conditions and no in-state restriction: the Neumann condition $\dot{h}|_0 = 0$ (their Eq. (30)) and the curvature-matching condition on the second derivative (their Eq. (59) in covariant form), which for a de Sitter tensor mode reads $\ddot{h}_k|_0 = k^2 h_k|_0$ (their Fourier-space Eq. (64)); together these fix the secular (ghost) coefficients in terms of the oscillatory ones and deliver the *general* two-parameter Einstein-Λ tensor mode $h_k = A\,(1 - \mathrm{i}k\eta)\,\mathrm{e}^{\mathrm{i}k\eta} + B\,(1 + \mathrm{i}k\eta)\,\mathrm{e}^{-\mathrm{i}k\eta}$, their Eq. (67). In the proxy frame the translation is immediate:

- Eq. (30), $\dot{x} = 0$: the variable q_2, i.e. exactly Maldacena's effect E_σ of (9);
- Eqs. (59)/(64), $\ddot{x} = -\omega_1^2 x$ (healthy Einstein dispersion): the phase-space variable

$$O_2 \;=\; \ddot{x} + \omega_1^2 x \;=\; p_2 + \omega_1^2 q_1, \qquad \text{constrained to 0,} \tag{12}$$

  regularized as $E'_\tau = \exp(-O_2^2/2\tau^2)$. We adopt the proxy dispersion $\ddot{x} = -\omega_1^2 x$, which is the *ghost-killing* sign derived just below—not a free choice. HLZ's Fourier-space condition (their Eq. (64), $\ddot{h}_k|_0 = +k^2 h_k|_0$, confirmed against the published text) carries the opposite *literal* sign yet selects the same object: together with Neumann (their Eq. (30), $\dot{h}|_0 = 0$) it uniquely carves out the Einstein two-parameter family $h_k = A(1-\mathrm{i}k\eta)\mathrm{e}^{\mathrm{i}k\eta} + B(1+\mathrm{i}k\eta)\mathrm{e}^{-\mathrm{i}k\eta}$ (their Eq. (67)). At the future boundary this family has *positive curvature*, $\ddot{h}(0) = +k^2 h(0)$ (locally $h \simeq 1 + \frac{1}{2}k^2\eta^2$)—the boundary signature the second condition enforces—while the corresponding healthy mode of the *flat* proxy is the oscillator $\mathrm{e}^{\pm\mathrm{i}\omega_1 t}$, whose curvature is $\ddot{x} = -\omega_1^2 x$. It is this boundary-curvature sign that inverts between the conformal-time and

flat-proxy descriptions—a property of the declared proxy, not a frame convention—so the literal $+k^2$ becomes $-\omega_1^2$; the proxy sign is fixed by ghost-killing (below). In normal-mode language $O_2 = 0$ is precisely "kill the ghost coordinate": for $x = y_1 + y_2$ with $\ddot{y}_i = -\omega_i^2 y_i$, the condition $\ddot{x} = -\omega_1^2 x$ forces $(\omega_1^2 - \omega_2^2)\, y_2 = 0$.

The crucial structural fact is that p_2 is the Ostrogradski momentum conjugate to q_2:

$$[q_2, O_2] = [q_2, p_2] = \mathrm{i}. \tag{13}$$

Maldacena's $\mathcal{M}$ fixes one quadrature of the velocity sector; the HLZ $\mathcal{M}'$ fixes a *canonically conjugate pair*. $\mathcal{M}'$ is therefore a joint position–momentum effect, and Heisenberg limits it: neither natural ordering of the two Gaussian conditions can beat the minimal phase-space cell. We price $\mathcal{M}'$ against the same reference vacuum Ψ_0 as $\mathcal{M}$, so the two effects are scored on identical input; HLZ's dropping of the in-state restriction constrains the *delivered* state, not the reference against which survival is measured.

Proposition 4.1 (Second-condition statistics)**.** *On the vacuum of Proposition 2.1,*

$$O_2\Psi_0 = \big(\mathrm{i}\,\delta\, q_2 + \omega_1 \delta\, q_1\big)\Psi_0, \qquad \mathrm{Var}_{\Psi_0}(O_2) = \frac{\delta}{2}\,\frac{\omega_1+\omega_2}{\omega_2} \xrightarrow{\delta\to 0} \delta, \tag{14}$$

and the symmetrized covariance vanishes exactly: $\mathrm{Re}\,\langle q_2 O_2\rangle_{\Psi_0} = 0$ *(indeed* $\langle q_2 O_2\rangle_{\Psi_0} = \mathrm{i}/2$*, pure commutator). Consequently the* (q_2, O_2) *Wigner marginal of* Ψ_0 *is an uncorrelated Gaussian, so the* Weyl*-ordered survival factorizes as* $p_1 p_2$*, with*

$$p_2(\tau,\delta) = \Big[1 + \frac{\mathrm{Var}(O_2)}{\tau^2}\Big]^{-1/2} \xrightarrow{\delta\to 0} 1. \tag{15}$$

Operator orderings that respect $[q_2, O_2] = \mathrm{i}$ *(product/Kraus) do* not *factorize; that non-factorization is precisely the* $1/(4\sigma^2\tau^2)$ *Heisenberg surcharge of Proposition 4.2.*

Proof. $p_2\Psi_0 = -\mathrm{i}\partial_{q_2}\Psi_0 = (\mathrm{i}bq_2 - cq_1)\Psi_0$ and $\omega_1^2 - c = \omega_1(\omega_1 - \omega_2) = \omega_1\delta$, giving the first identity. Then $\mathrm{Var}(O_2) = b^2\langle q_2^2\rangle + \omega_1^2\delta^2\langle q_1^2\rangle = \frac{b}{2} + \frac{\omega_1^2\delta^2}{2\alpha} = \frac{\delta}{2}\big(1 + \frac{\omega_1}{\omega_2}\big)$ (the cross terms cancel between the $\pm\mathrm{i}$ pieces), and $\langle q_2 O_2\rangle = \mathrm{i}b\langle q_2^2\rangle = \mathrm{i}/2$. Since Ψ_0 is Gaussian, O_2 is Gaussian-distributed and (15) is the standard Gaussian-effect integral. Grid values (spectral p_2, 2048-point FFT) reproduce (14) to five digits at $\delta = 0.5, 0.1, 0.02$ and the covariance to 10^{-18} at $\delta = 0.3, 0.05$. □

Equation (14) says the prepared vacuum *already nearly satisfies* the second HLZ condition as $\delta \to 0$: the second condition is asymptotically free in scale. All the δ-collapse continues to come from the Neumann factor. This conclusion—and with it the equal-leading-power verdict of Proposition 4.2—rests on the sign $O_2 = p_2 + \omega_1^2 q_1$, which is the ghost-killing sign of (12) and *also* the finite-variance one: the opposite sign $O_2 = p_2 - \omega_1^2 q_1$ isolates no proxy mode and gives $\mathrm{Var}(O_2) = \frac{\delta}{2} + \omega_1(\omega_1+\omega_2)^2/(2\omega_2\delta) \to \infty$ (leading power 1, not $\frac{1}{2}$). Ghost-killing and finite variance therefore select the same sign, and HLZ's Eq. (64)—$\ddot{h}_k|_0 = +k^2 h_k|_0$, confirmed against the published text—is consistent with it under the role inversion of (12). The second-condition sign is thus derived; the only residual modeling input is the declared proxy dictionary itself (Section 8).

4.2 The price of $\mathcal{M}'$

Because the two constrained variables are conjugate, the survival of $\mathcal{M}'$ depends on the operator ordering of the two Gaussian factors—an $O(1)$, not power-law, effect.

Proposition 4.2 ($\mathcal{M}'$ price sheet)**.** *Let $p_1 = p(\sigma, \delta)$ of (10) and $p_2 = p_2(\tau, \delta)$ of (15). Then:*

(i) (Weyl ordering; exact) The operator with Weyl symbol $\exp(-q_2^2/2\sigma^2 - O_2^2/2\tau^2)$ has vacuum expectation exactly $p_1 p_2$; it is a bona fide effect (positivity $E \geq 0$) iff $\sigma\tau \geq 1/2$. Here $\sigma\tau \geq 1/2$ is the Gaussian positivity/uncertainty condition securing the lower bound $E \geq 0$; the upper bound $E \leq 1$ then follows because at and above the minimal cell the operator is, up to a metaplectic (Gaussian-unitary) conjugation, a single-mode Gaussian effect whose Fock eigenvalues lie in $[0, 1)$, reducing to $\mathbf{1}$ only as $\sigma, \tau \to \infty$.

(ii) (Product/Kraus ordering; exact) The always-legitimate effect $E = K^\dagger K$ with $K = \mathrm{e}^{-O_2^2/4\tau^2}\, \mathrm{e}^{-q_2^2/4\sigma^2}$ has vacuum survival given exactly by the closed form

$$p_{\mathcal{M}'} \;=\; p_1 \Big[1 + \frac{\mathrm{Var}(O_2)}{\tau^2} + \frac{1}{4\sigma^2\tau^2}\Big]^{-1/2}, \tag{16}$$

whose sub-minimal-cell (sharp-effect) limit, taken along the joint $\sigma, \tau \to 0$ path (so that the Heisenberg term $1/(4\sigma^2\tau^2)$ dominates the bracket, i.e. $\sigma\tau \to 0$), is exactly $p_{\mathcal{M}'} \to 2\sigma\tau\, p_1$: the Maldacena survival times the phase-space cell $2\sigma\tau$ in minimal-uncertainty units. The extra $1/(4\sigma^2\tau^2)$ in the bracket—absent from the Weyl product $p_1 p_2$ of (i)—is the Heisenberg back-reaction of squeezing q_2 onto its conjugate $p_2 \subset O_2$, i.e. the $O(1)$ ordering surcharge.

(iii) Both orderings have the same leading detuning power, exactly $1/2$: $p_{\mathcal{M}'} \sim \sigma\sqrt{2\delta}$ as $\delta \to 0$ up to $O(1)$ prefactors, since $p_1 \to \sigma\sqrt{2\delta}$ while the bracket in (16) tends to a δ-independent constant (recall $\mathrm{Var}(O_2) \to 0$). The $\delta \to 0$ leading power of the closed form (16) is therefore $1/2$ for every (σ, τ).

Proof. (i) The expectation of a Weyl-ordered operator is the phase-space average of its symbol against the Wigner function; the Wigner function of the Gaussian Ψ_0, marginalized to the (q_2, O_2) plane, is an uncorrelated Gaussian with the variances of Proposition 4.1, so the double Gaussian integral factorizes into $p_1 p_2$. Positivity of the Weyl quantization of a Gaussian symbol holds iff the symbol's covariance admits the minimal-cell bound, i.e. $\sigma\tau \geq 1/2$. The two single-factor limits are exact on the grid: at $\delta = 0.1$, $\tau = 10^3$ returns p_1 and $\sigma = 10^3$ returns p_2, each to five digits.

(ii)–(iii) In the product ordering $\langle\Psi_0|K^\dagger K|\Psi_0\rangle = \langle\phi|\mathrm{e}^{-O_2^2/2\tau^2}|\phi\rangle$ with $\phi = \mathrm{e}^{-q_2^2/4\sigma^2}\Psi_0$, still Gaussian. Three steps. (1) The norm ratio is Maldacena's survival exactly, $\|\phi\|^2 / \|\Psi_0\|^2 = [1 + 1/(2\sigma^2\delta)]^{-1/2} = p_1$, since $|\phi|^2 = \mathrm{e}^{-q_2^2/2\sigma^2}|\Psi_0|^2$ reproduces the integral of Proposition 3.1. (2) In the normalized state $\phi/\|\phi\|$, O_2 is again a zero-mean Gaussian, now with variance $V' = \mathrm{Var}_{\Psi_0}(O_2) + 1/(4\sigma^2)$: narrowing q_2 by the factor $\mathrm{e}^{-q_2^2/4\sigma^2}$ broadens the conjugate $p_2 \subset O_2$ by exactly $1/(4\sigma^2)$, the Heisenberg back-reaction of squeezing q_2. (3) The Gaussian-effect overlap then gives $\langle\phi|\mathrm{e}^{-O_2^2/2\tau^2}|\phi\rangle / \|\phi\|^2 = [1 + V'/\tau^2]^{-1/2}$. Multiplying (1) and (3) yields (16) exactly. Along the joint $\sigma, \tau \to 0$ path (so that $\sigma^2\mathrm{Var}(O_2) \to 0$) the $1/(4\sigma^2\tau^2)$ term dominates the bracket, giving $p_{\mathcal{M}'} \to 2\sigma\tau\, p_1$; and since $\mathrm{Var}(O_2) \to 0$ with $p_1 \to \sigma\sqrt{2\delta}$, the detuning exponent is exactly $1/2$. A high-resolution grid tracks (16) (with grid-resolution artifacts at the smallest δ, where the vacuum width $\sim \delta^{-1/2}$ outruns the box; the tabulated values are the analytic closed form); the comparison table follows. □

δ	p_1	p_2	$p_1 p_2$ (Weyl)	$p_{\mathcal{M}'}$ (16)	$2\sigma\tau\, p_1$
0.500	0.09950	0.11471	0.01141	0.00196	0.00199
0.100	0.04468	0.29417	0.01314	0.00089	0.00089
0.020	0.02000	0.57540	0.01151	0.00040	0.00040
0.004	0.00894	0.84491	0.00756	0.00018	0.00018

Table 2: Pricing $\mathcal{M}'$ at $\sigma = \tau = 0.1$ (so $2\sigma\tau = 0.02$). The Neumann factor p_1 carries all the δ-collapse; the second-condition factor $p_2 \to 1$ (asymptotically free); the product-ordered survival is the exact closed form (16), which in this deep sub-cell (sharp-effect) regime tracks $2\sigma\tau\, p_1$, i.e. Maldacena's price times a Heisenberg-cell surcharge. Here $\sigma\tau = 0.01 < 1/2$, so by Proposition 4.2(i) the Weyl operator is not positive in this regime; the displayed $p_1 p_2$ (Weyl) column is therefore the expectation of a non-effect—a formal comparator, not a survival probability.

4.3 What the surcharge buys

The fidelity scorecard exposes the qualitative difference between the two exorcisms. On the same grids, at $\sigma = \tau = 0.05$:

δ	$F_{\mathcal{M}}$ (Neumann only)	$F_{\mathcal{M}'}$ (both conditions)
0.500	0.7996	0.0635
0.100	0.5494	0.0506
0.020	0.2744	0.0508
0.004	0.1258	0.0786

$F_{\mathcal{M}}$ tracks the closed form (11); $F_{\mathcal{M}'}$ is roughly flat and small, and this is *by design*, not by failure: $\mathcal{M}'$ does not target the pure ω_1 vacuum at all. It targets the general two-parameter (A, B) Einstein solution with no in-state preparation—a mixed delivered state whose overlap with any single pure vacuum is correspondingly reduced (small in the numerics above, not by a general bound). Pure-vacuum fidelity is the right scorecard for $\mathcal{M}$ and the wrong one for $\mathcal{M}'$.

The comparison is then clean:

- *Survival:* $p_{\mathcal{M}} \sim \sigma\sqrt{2\delta}$ and $p_{\mathcal{M}'} \sim \sigma\sqrt{2\delta}$ up to $O(1)$: the second HLZ condition adds only a Heisenberg surcharge (it fixes $\ddot{x}$, conjugate to the $\dot{x}$ that Neumann fixes) and is not a new source of collapse, because $\mathrm{Var}(O_2) \sim \delta$ already vanishes.
- *Delivery:* $\mathcal{M}$ delivers (at $\sqrt{\delta}$-collapsing fidelity) a pure but over-broad vacuum; $\mathcal{M}'$ delivers the general Einstein solution.

This is HLZ's own claim—Neumann alone yields no general solution—re-expressed as a price: the extra condition buys *generality*, not economy, and the conformal ($\delta \to 0$) collapse of the survival is not evaded by either scheme. In the currency of this series, there is no cheap exorcism of the conformal graviton pair by future boundary data: the price is set by the exceptional-point degeneration of the state, which no rearrangement of the boundary conditions removes.

5 The pole sheet is readable in the divisor

We now turn from boundary-condition exorcisms to propagator-prescription exorcisms, and to the spectral question underneath all of them: what, exactly, distinguishes a ghost from an ordinary unstable particle with the same complex pole pair? Buoninfante [6, 8] gives the sharp QFT answer: only the *sheet*. Both have poles at $m^2 \pm im\Gamma$; the unstable particle's pair sits on the second (unphysical) sheet and its Källén–Lehmann representation is a positive density on the real axis with sum rule $1 = C$ (complete norm transfer to the continuum: the particle decays) [8, Eq. (34)]; the ghost's pair sits on the first (physical) sheet and the spectral decomposition contains, in addition to a *still-positive* continuum, discrete negative-residue complex-pole terms $-Z\Delta_F(m^2+im\Gamma) - Z^*\Delta_F(m^2-im\Gamma)$ [8, Eq. (40)], with sum rule $Z+Z^* = 1+C$: the one-particle component persists [8, Eq. (44)]. We emphasize the precise locus of positivity failure, because it is easy to state wrongly: the ghost's continuum is positive; what fails is positivity of the *total* spectral measure, through the discrete complex-pole part. The two survival regimes arise from two distinct mechanisms, and it is worth keeping them separate. A positive spectral measure on the real axis gives, by Bochner's theorem, a *bounded* amplitude $|A(t)| \le 1$—the second-sheet resonance. What makes the first-sheet ghost's $|A(t)|$ *unbounded* is not sign-indefiniteness as such—a finite signed measure on the real axis still gives a bounded $|A(t)|$—but that its spectral support includes *complex* energies $m \pm i\gamma$ off the real axis, whose $e^{\pm\gamma t}$ legs make $|A(t)|$ grow, as the divisor will now report.

5.1 The Bochner discriminant

Proposition 5.1 (Pole sheet → process class)**.** *Let $A(t)$ be the survival amplitude of a single mode.*

(i) (Second sheet; rigorous.) If $A(t) = \int e^{-iEt}\, d\mu(E)$ for a positive probability measure μ on $\mathbb{R}$—the case of an ordinary unstable particle, whose dressed propagator has a positive spectral density—then A is a positive-definite function and Bochner's theorem gives $|A(t)| \le A(0) = 1$ for all t. The two-state survival matrix

$$\Gamma(t) = \begin{pmatrix} |A(t)|^2 & 0 \\ 1 - |A(t)|^2 & 1 \end{pmatrix}$$

is then column-stochastic with no conditioning*: because $|A| \le 1$, an unconditioned substochastic survival process exists on the second sheet whatever the negativity pattern. Divisor negativity over $[t_1, t_2]$ occurs iff $|A(t_2)| > |A(t_1)|$ (net increase over the interval; backflow); its magnitude $M_{21} = 1 - |A(t_2)|^2/|A(t_1)|^2$ is bounded only where* $\inf|A| > 0$ *and diverges near an amplitude null. For absolutely continuous μ the Riemann–Lebesgue lemma sends $A(t) \to 0$ in amplitude, so the negativity is transient* when $|A|$ is eventually monotone *(as for the smooth density below)—the transient-indivisibility class of [1]; a compactly supported or oscillatory a.c. measure (μ uniform on an interval, $A \propto \mathrm{sinc}$) instead revives backflow at every amplitude null. A measure with atoms likewise stays unconditioned ($|A| \le 1$) without being transient—a single atom over an a.c. background can even make $|A|$ eventually monotone* increasing *($\mu = 0.7\,\delta_0 + \rho$ gives $A = 0.7 - 0.2e^{-t} + 0.5e^{-2t}$, permanent second-sheet negativity with no conditioning). What is invariant is the dichotomy by* boundedness of *$|A|$, not the negativity pattern: $|A| \le 1$ always admits an unconditioned process, while the unbounded $|A|$ of (ii) forces conditioning.*

(ii) *(First sheet; mode-level analytic, closed forms numerically re-verified.) If the mode dynamics carries a first-sheet complex-conjugate pair—eigenfrequencies* $m \pm \mathrm{i}\gamma$, $\gamma > 0$, *the mode realization of Buoninfante's ghost pair—then no positive spectral measure exists for* A, *and indeed* $|A(t)|$ *grows like* $\cosh(\gamma t)$ *on the configuration basis (any beable state overlapping the growing leg): since* $|A(t)| > 1$ *at finite* t, no *unconditioned sub-stochastic survival process exists at any coarse-graining. The survival-conditioned dilation* (2) *is mandatory, and for the dimer realization* (3) *it lands in the terminal class of [1]: permanent divisor negativity past* t_c *of* (4), *with the late-time conditioned generator pinned at the floor* $-(a + \widetilde{k})$.

Proof. (i) is Bochner's theorem plus inspection: $|A(t)| = |\int \mathrm{e}^{-\mathrm{i}Et}\mathrm{d}\mu| \le \int \mathrm{d}\mu = 1$; column sums of $\Gamma(t)$ are 1; the divisor (1) of the survival pair has entries $M_{11} = |A(t_2)|^2/|A(t_1)|^2$ and $M_{21} = 1 - M_{11}$, negative iff $|A|$ increased; Riemann–Lebesgue sends $A \to 0$ in amplitude, so the revivals decay when $|A|$ is eventually monotone (but not for a compactly supported or oscillatory a.c. measure, whose $|A|$ nulls revive backflow). A quantitative illustration: for the positive density $\rho(E) \propto \sqrt{E}\,(1+E)^{-1/2}\big[(E-m_0)^2 + \Gamma_0^2/4\big]^{-1}$ ($m_0 = 3$, $\Gamma_0 = 0.4$), the computed $\max_t |A(t)| = 1.000000$, and the residual backflow episodes are at the 10^{-9} level (numerical noise), with no growing revival—this particular density is essentially monotone; the general argument allows densities with sharper thresholds to show visible but decaying revivals, though the public script tests only the smooth density above. (ii) The growth statement is elementary: with eigenfrequencies $m \pm \mathrm{i}\gamma$ the evolution has singular values $\mathrm{e}^{\pm\gamma t}$ up to bounded factors, so $\|U(t)\| \ge C\mathrm{e}^{\gamma t}$; on the dimer (3) this is exact via (5). The computed $\max_{t\le 6} \left\|\mathrm{e}^{-\mathrm{i}Ht}\right\| = 1098.9$ at $(a, \kappa) = (1.5, 1)$. The terminal-class closed forms (4) are quoted from [1] and re-verified here: permanent-negativity onset 0.6400 on a 0.005 time grid against the closed form 0.6487, and late-time generator off-diagonal -2.6031 against the floor $-(a+\widetilde{k}) = -2.6180$ (both discrepancies are finite-difference artifacts that shrink with the step). We state part (ii) at mode level: the assertion that the *field-theoretic* ghost two-point function has exactly this discrete negative-residue structure is Buoninfante's [8, Eqs. (40),(44)], not re-derived here. □

Remark 5.2 (The classifier crosses at the sheet boundary). Sweeping a two-mode model with eigenvalues $3 + \mathrm{i}g_{\mathrm{eff}}$ and $3 - 0.4\mathrm{i}$ across the boundary: for $g_{\mathrm{eff}} = -0.4, -0.1, 0.0$ the maximal evolution norm is 1.000 (unconditioned, resonance class); at $g_{\mathrm{eff}} = +0.1$ it is 20.1 and at $+0.4$ it is 1.6×10^5 (conditioning mandatory; not terminal—this sweep is decoupled). The process-class classifier changes exactly at the sheet boundary $g_{\mathrm{eff}} = 0$—there is no intermediate regime.

The slogan form of Proposition 5.1:

> *Unstable particles survive unconditioned; ghosts survive only conditioned. The pole sheet is readable in the divisor by the* boundedness *of the amplitude: an unconditioned process on the second sheet (*$|A| \le 1$*), and—for the dimer—terminal negativity with a generator floor on the first (*$|A|$ *unbounded).*

This gives the divisor currency an operational meaning directly in QFT terms: handed only the survival record of a mode, one can read off which sheet its pole pair inhabits—from whether $|A|$ stays bounded (second sheet) or grows without bound (first). The reading is a converse only within the two-class family above, and for states overlapping the growing leg: a signed real-axis measure also keeps $|A|$ bounded, and a first-sheet mode probed in a decaying-leg-only state leaves a bounded record.

5.2 The masking law is the terminal-class dimer's cross-population

Buoninfante's second and third papers establish what he calls *anti-instability*: in the indefinite-metric theory the ghost neither decays (norm conservation forbids negative-norm → positive-norm transfer) nor runs away; instead the would-be decay manifests as *multi-particle masking*, an overlap with multi-particle states that grows as $\sinh^2(\Gamma t/2)$: the general overlap is his [7, Eq. (84)], whose at-rest ($\vec{p} = 0$) narrow-width form [7, Eq. (88)] is exactly $\sinh^2(\Gamma t/2) + O(\Gamma^2)$, on the timescale $2/\Gamma$; no free asymptotic one-particle ghost state exists, and the Feynman ghost *appears* to retain a single arrow of causality on shell—the source's own hedge, which flags that first-sheet complex-conjugate poles could still induce violations.

Proposition 5.3 (Masking law = cross-population growth)**.** *For the broken dimer* (3)*, the cross-population is exactly*

$$|U_{12}(t)|^2 = \frac{\kappa^2}{\widetilde{k}^2} \sinh^2(\widetilde{k}t),$$

by (5)*. Setting* $\widetilde{k} = \Gamma/2$ *this is Buoninfante's masking law* $\sinh^2(\Gamma t/2)$ *up to the constant prefactor* $(\kappa/\widetilde{k})^2$*, including the timescale* $2/\Gamma$*. (The identity is exact—algebraic, not fitted.)*

The identification says the two formalisms are computing the same growth and booking it differently. In Buoninfante's closed, indefinite-metric currency, nothing decays and nothing runs away: the growth is an overlap, and "anti-instability" is the statement that the indefinite norm is conserved. In the beable currency of this series, the same growth is the reason no unconditioned process exists, and conditioning divides it out exactly: *after* conditioning, nothing physical grows—which is the conditioned transcription of anti-instability. Two bookkeepings, one pole pair, one shared closed form. We are careful about scope: Buoninfante's masking is an overlap statement in indefinite-metric QFT; our reading adds the sample-space price tag and claims nothing about his finite-time resummation—what this paper adds is the currency, not a new field theory.

6 A taxonomy of exorcisms and their currencies

Buoninfante's first paper organizes the propagator prescriptions for a ghost resonance into an $(a; b, c)$ taxonomy [6, Eqs. (5)–(6)]: $a = -1$ selects the ghost (first sheet) and $a = +1$ the non-ghost/unstable particle (second sheet, cf. Section 5), and (b, c) select how the complex pair is routed through the contour. The three standard exorcisms, and the unstable-particle control, then line up against the trilemma of [3]—a coupled-ghost theory keeps at most two of {coupled ghosts, empirical Lorentz invariance, definite configurations}—as follows.

Three remarks, in increasing order of contentiousness.

First, the Feynman ghost row is Proposition 5.1(ii) verbatim: keeping the ghost coupled, causal, and on the first sheet forfeits the unconditioned sample space; the theory survives in the beable currency only as a survival-conditioned terminal-class process. This is the "exit-the-sample-space" horn of the trilemma, now anchored in the sheet structure rather than asserted.

Second, the anti-Feynman row is structurally the Donoghue–Menezes route [10]: a positive-norm *asymptotic spectrum* and a unitary S-matrix are retained at the cost of microcausality (their unstable-ghost loops arrow the ghost propagation backwards). We note explicitly that this alignment is ours to draw: Buoninfante's trilogy does not itself connect the $(b = c = -1)$ contour routing to the "Merlin" treatment of [10]; the alignment is between that routing and

Prescription	Norms	Causal structure	Process reading	Currency paid
Feynman ghost ($b = c = +1$)	indefinite (odd-ghost < 0)	causal	conditioning mandatory, terminal class (§5)	sample space (no beable seat)
anti-Feynman ($b = c = -1$)	positive	acausal (reversed arrow)	unconditioned, reversed arrow	causal order / Lorentz
fakeon ($b = -c$)	—	purely virtual (Im $= 0$)	no asymptotic state to condition	silence (excision)
unstable particle ($a = +1$, 2nd sheet)	positive	causal	unconditioned sub-stochastic	none

Table 3: The $(a; b, c)$ prescription taxonomy of [6] mapped to the trilemma currencies of [3]. Every exorcism pays in at least one currency; only the genuine second-sheet unstable particle is free.

the acausal-but-unitary corner they occupy, and it slots into the Lorentz/causal-order column of the trilemma table (the currency whose empirical price [2] computes for coupled-ghost vacuum decay).

Third, the fakeon row is Anselmi's, and here sits the live dispute. Anselmi, Briscese, Calcagni, and Modesto [9] argue that among the prescriptions for complex poles only the fakeon is viable; Buoninfante's anti-instability result argues the Feynman ghost is consistent as it stands [7, 8]. Read in Table 3, the two camps are not disagreeing about a computation—they are pricing in different currencies. The fakeon buys consistency with silence: the ghost never appears as an asymptotic state, at the price of excising it from the sample space altogether (there is nothing left to condition on). Buoninfante's Feynman ghost buys consistency with conditioning: the ghost persists (sum rule $Z + Z^* = 1 + C$), is masked rather than decayed, and—in our transcription—its beable process is survival-conditioned, terminal class. Each camp's "viability" is consistency in its own currency. The trilemma does not adjudicate which currency is the right one to pay; it prices both, and this paper's contribution is that the *boundary-condition* prices are now numbers (Sections 3–5) rather than slogans.

7 Relation to final-state post-selection

The general idea that a future boundary condition is a post-selection is not ours. It is the engine of the Horowitz–Maldacena black-hole final-state proposal [11], where a maximally entangled final state at the singularity post-selects the interior dynamics; and it has been proposed in cosmology, where Anastopoulos reads cosmic acceleration itself as a quantum post-selection effect [12]. The two-state-vector and decoherent-histories traditions provide the general formalism. What we have not found anywhere is the *intersection* with ghost removal: no reading of Maldacena's Neumann condition [4] (or the HLZ variant [5]) as a survival-conditioned quantum effect, and no computation of the survival/fidelity price of any ghost-removing boundary condition. Our scope claims are therefore limited to exactly that intersection: (i) the identification of ghost-removal boundary conditions as effects in the technical sense, with the state-prep/effect split of Section 3.1; and (ii) the quantitative price sheet, Tables 1 and 2, with its conformal collapse. Likewise, on the prescription side, the pole-sheet/process-class dictionary of Proposition 5.1 and the masking identification of Proposition 5.3 are, to our knowledge, new, while the spectral inputs are entirely Buoninfante's [6, 7, 8].

8 Limitations

Every caveat below is substantive; none is decorative.

1. *The sharp effect is improper.* The exact Neumann projection has survival probability zero; all prices are stated for the regularized Gaussian family (9) at fixed σ, with the $\sigma \to 0$ asymptotics displayed. The δ-scaling—the physical content—is uniform in σ.

2. *The quantization is a declared choice.* We price the Dirac-normalizable (beable-carrying) quantization of the PU mode (Proposition 2.1). The contour-rotated Bender–Mannheim quantization [14] is a different leg of the taxonomy, and the price of the same boundary condition there is a separate (open) computation; the answer may differ.

3. *de Sitter only; Minkowski is the singular limit.* The proxy's clean graviton $\otimes$ partner split exists only at $\delta > 0$; flat space is the Jordan-block point where the effect $\mathcal{M} = \mathbf{1} \otimes E$ is not even well defined. We treat $\delta \to 0$ as the singular limit and verify that the price sheet detects it (Corollary 3.3)—that the currency diagnoses the obstruction is a feature, but it also bounds the method: nothing here prices a flat-space exorcism, because on this analysis there is no well-posed one to price.

4. *No double-counting, and the split is a modeling decision.* We count positive frequency as preparation and Neumann as the effect. This matches the logical roles in [4], but a reader who insists on counting the in-state restriction as part of the removal would obtain a different (strictly smaller) survival; our convention is the charitable one.

5. *Tree level only.* The whole analysis, like the trilemma it extends [3], is tree-level. Beyond tree level a future-boundary effect inherits the known interaction obstruction of final-state proposals [11]—interactions between the post-selected sector and spectators degrade the conditioning—and we make no claim there.

6. *The proxy is declared, not derived.* The detuned PU mode with δ standing for the $\Delta = 3/\Delta = 2$ splitting is a mode-level translation of [4], adopted for its exact solvability. Its qualitative structure (four solutions, the t-growing partner, the Jordan degeneration at flat space) matches the graviton system, and its prediction reproduces the HLZ flat-space obstruction; but the numerical prefactors in Tables 1–2 are properties of the proxy, not of conformal gravity. In particular, where [4]'s construction is an exact tree-level identity, the proxy delivers a fidelity $F < 1$ at $O(1)$ δ (e.g. $F \approx 0.80$ at $\delta = 0.5$): the proxy has no counterpart of the renormalization step that makes that identity exact, so the $O(1)$ part of the deficit is a proxy artifact. Moreover the $\delta \to 0$ collapse of the *fidelity* specifically is reference-convention-dependent (a co-scaling Einstein reference removes it; text after Proposition 3.2); the convention-free claimed content is the survival collapse $p \sim \sigma\sqrt{2\delta}$ and the delivered state's absolute width divergence $\alpha^{-1/2} \sim \delta^{-1/2}$.

7. *One result is a numerical re-verification of closed forms.* The terminal-class landing of the conditioned first-sheet process uses the critical time and generator floor proved in closed form in [1], re-verified here numerically; it is presented at exactly that strength in Proposition 5.1.

9 Conclusion

Every exorcism has a price, and the divisor currency makes it legible.

Maldacena's future boundary condition, the cleanest ghost removal in the literature, is per mode a Gaussian survival effect on a beable variable, and its full price sheet is closed form: survival $\sigma\sqrt{2\delta}$, Einstein fidelity $2\sqrt{\omega_2\delta}$ (in units $\omega_1 = 1$, against a fixed Einstein reference), joint figure of merit linear in the de Sitter splitting. The exorcism is affordable exactly where Maldacena performed it—in de Sitter, at $\delta = O(1)$—and its price diverges at the conformal point, which is the flat-space obstruction of Hell, Lüst, and Zoupanos rediscovered as a collapse of the price sheet. Their own two-condition variant fixes a canonically conjugate pair and is therefore Heisenberg-limited: same leading price, an $O(1)$ surcharge, and what the surcharge buys is generality (the full Einstein solution space) rather than economy. There is no cheap version: the cost is set by the exceptional-point degeneration of the state itself.

On the propagator side, the pole sheet is readable in the divisor. Positive spectral measure—the second-sheet unstable particle—gives $|A(t)| \leq 1$ by Bochner's theorem and an unconditioned sub-stochastic process whose divisor negativity decays when the amplitude is eventually monotone decreasing (transient-indivisibility class). The first-sheet ghost pair, whose spectral decomposition carries discrete negative-residue complex poles with the persistence sum rule $Z + Z^* = 1 + C$, admits no unconditioned process at all: conditioning is mandatory; for the dimer realization it is terminal, with the critical time and generator floor of the first companion paper. Buoninfante's masking law is the *time dependence* of the terminal-class dimer's cross-population—the pre-conditioning growth that conditioning divides out—up to the normalization $(\kappa/\widetilde{k})^2$; his anti-instability and our conditioning are the closed-currency and beable-currency descriptions of one growth. And the prescription taxonomy—Feynman ghost, anti-Feynman, fakeon—pays out in the three trilemma currencies: sample space, causal order, silence. The Anselmi–Buoninfante dispute over which prescription is viable is a dispute over which currency one is willing to pay in; both sides are priced here.

The unifying statement is the one the title promises. A ghost that is removed has not been annihilated; it has been *paid for*—conditioned on survival of a boundary effect, or bought out by a contour choice or an excision—and the price always shows up in the books: as a vanishing survival probability, as a fidelity collapse, as an abandoned causal order, or as a sample space with one fewer inhabitant. Unstable particles survive unconditioned. Ghosts survive only by paying—conditioned, or in another trilemma currency. For the boundary-condition exorcisms priced here, that price is now a number.

Author contributions and use of AI

This manuscript was produced through a collaboration, directed by the author, with the large language model Claude Fable 5 (Anthropic), which carried out the technical work: the research program, the derivations and proofs, the numerical experiments, the literature analysis, and the text, including this statement. J.T. posed the research direction, provided iterative review, and takes responsibility for the decision to circulate. The research notebook in the program's public repository records the development of this paper's results, including the scoping analyses in which the state-preparation/effect split and the Heisenberg-conjugate structure of the two-condition scheme were identified before the prices were computed.

Code availability

All numerical claims are reproducible from three standalone scripts (NumPy/SciPy, seconds to minutes each, controls printed inline) in the paper's repository:

https://github.com/embeddetech/p08-exorcised-ghosts

(also archived, with the full program history, in the hub repository https://github.com/embeddetech/ghosts-program):

- `scripts/p8_maldacena_price.py` (Section 3): vacuum verification, survival and fidelity closed forms, the price sheet;
- `scripts/p8_hlz_mprime.py` (Section 4): $\mathrm{Var}(O_2)$, covariance, ordering-dependent survival, delivered-state fidelities;
- `scripts/p8_buoninfante_map.py` (Section 5): the Bochner bound, the terminal-class closed forms, the sheet-crossing sweep.

References

[1] J. Torkelson, *Indivisible stochastic processes for non-Hermitian and higher-derivative quantum systems: detailed balance, broken $\mathcal{PT}$ symmetry, and the explosion of ghosts*, companion paper (2026), https://github.com/embeddetech/p01-indivisible-ghosts.

[2] J. Torkelson, *Ghost vacuum decay as a stochastic process: instantaneous states, the boost-orbit measure, and the foliation that ghosts make visible*, companion paper (2026), https://github.com/embeddetech/p03-instantaneous-ghosts.

[3] J. Torkelson, *Conformal gravity must choose: exceptional-point kinematics, vacuum decay, and the sample space of Weyl gravity*, companion paper (2026), https://github.com/embeddetech/p05-choosing-ghosts.

[4] J. Maldacena, *Einstein gravity from conformal gravity*, arXiv:1105.5632 (2011).

[5] A. Hell, D. Lüst, and G. Zoupanos, *On the ghost problem of conformal gravity*, JHEP **08** (2023) 168, arXiv:2306.13714.

[6] L. Buoninfante, *Remarks on ghost resonances*, JHEP **02** (2025) 186, arXiv:2501.04097.

[7] L. Buoninfante, *Asymptotic quantum dynamics of ghost fields*, arXiv:2605.29047 (2026).

[8] L. Buoninfante, *Ghosts versus unstable particles in quantum field theory*, arXiv:2606.18349 (2026).

[9] D. Anselmi, F. Briscese, G. Calcagni, and L. Modesto, *Amplitude prescriptions in field theories with complex poles*, JHEP **05** (2025) 145, arXiv:2503.01841.

[10] J. F. Donoghue and G. Menezes, *Unitarity, stability, and loops of unstable ghosts*, Phys. Rev. D **100**, 105006 (2019), arXiv:1908.02416.

[11] G. T. Horowitz and J. Maldacena, *The black hole final state*, JHEP **02** (2004) 008, arXiv:hep-th/0310281.

[12] C. Anastopoulos, *Final states in quantum cosmology: cosmic acceleration as a quantum postselection effect*, Phys. Rev. D **111**, 064045 (2025), arXiv:2401.07662.

[13] A. Pais and G. E. Uhlenbeck, *On field theories with non-localized action*, Phys. Rev. **79**, 145 (1950).

[14] C. M. Bender and P. D. Mannheim, *No-ghost theorem for the fourth-order derivative Pais–Uhlenbeck oscillator model*, Phys. Rev. Lett. **100**, 110402 (2008), arXiv:0706.0207.